

Configuring Networking

Networking Fundamentals

- **IPv4 addresses:** These are based on 32-bit addresses and have four octets, separated by dots, such as 192.168.10.100.

- **IPv6 addresses:** These are based on 128-bit addresses and are written in eight groups of hexadecimal numbers that are 16 bits each and separated by colons. An IPv6 address may look like fe80:badb:abe01:45bc:34ad:1313:6723:8798.

The network addresses can be assigned in two ways:

- **Fixed IP addresses:** Useful for servers that always need to be available at the same IP address.

- **Dynamically assigned IP addresses:** Useful for end users' devices, and for instances in a cloud environment. To dynamically assign IP addresses, a Dynamic Host Configuration Protocol (DHCP) server is usually used.

Names for network cards:

In RHEL 8, the default names for network cards are based on firmware, device topology, and device types. This leads to network card names that always consist of the following parts:

- Ethernet interfaces begin with *en*, WLAN interfaces begin with *wl*, and WWAN interfaces begin with *ww*.
- The next part of the name represents the type of adapter. An *o* is used for onboard, *s* is for a hotplug slot, and *p* is for a PCI location. Administrators can also use *x* to create a device name that is based on the MAC address of the network card.
- Then follows a number, which is used to represent an index, ID, or port.
- If the fixed name cannot be determined, traditional names such as `eth0` are used.

Based on this information, device names such as `eno16777734` can be used, which stands for an onboard Ethernet device, with its unique index number.

Apart from this default device naming, network cards can be named based on the BIOS device name as well. In this naming scheme, names such as `em1` (embedded network card 1) or `p4p1` (which is PCI slot 4, port 1) can be used. To use this kind of naming, the `biosdevname` package must be installed.

Validating Network Configuration

- IP address and subnet mask
- Routing
- Availability of ports and services

Validating Network Address Configuration

- Use **ip addr** to configure and monitor network addresses.

- Use **ip route** to configure and monitor routing information.
- Use **ip link** to configure and monitor network link state.

Displaying IP addresses/Prefix:

```
[root@master ~]# ifconfig
```

```
[root@master ~]# ifconfig ens33
```

```
[root@master ~]# ip help
```

```
[root@master ~]# ip addr help
```

```
[root@master ~]# ip addr          (Display IP information)
```

```
[root@master ~]# ip addr show    (Display IP information)
```

```
[root@master ~]# ip addr show ens33
```

```
[root@master ~]# ip link show
```

```
[root@master ~]# ip -s link show
```

```
[root@master ~]# ip -s link show ens33
```

Displaying Routing (Gateway):

```
[root@master ~]#ip route
```

Or [root@master ~]# route -n

Display Name Server:

```
[root@master ~]#cat /etc/resolve.conf
```

Display all Network config. In one command:

```
[root@master ~]#nmcli device show ens192
```

Configure Network Configuration :

- GUI
- TUI (nmtui)
- CLI (nmcli)
- Editing in files.

Hint:

- A device is a network interface card.
- A connection is the configuration that is used on a device.

In RHEL 8, you can create multiple connections for a device. Instead of modify the current configuration based on one interface, it's make sense to create a connection (profile) and if you done with it (the profile) you can easily delete it.

Before configure the NW Configs, Let's ask this "Who can change NW configuration ?!! Normal Users or ROOT !!!"

Obviously, the root user can make modifications to current networking. However, if an ordinary user is logged in to the local console, this user is able to make changes to the network configuration as well. As long as the user is using the system keyboard to enter either a graphical console or a text-based console, these permissions are granted. This is because users are supposed to be able to connect their local system to a network. Notice that regular users that have used **ssh** to connect to a server are not allowed to change the network configuration. To check your current permissions, use the **nmcli gen permissions** command

Configuring Networking with nmtui:

```
[root@client ~]# nmtui
```

```
[root@client ~]# nmcli con down ens33 ; nmcli con up ens33
```

Nmcli con down == ifdown

Nmcli con up == ifup

Editing and adding connection with nmcli utility:

```
[root@client ~]# nmcli dev show <devicename>
```

```
[root@client ~]#nmcli dev connect/disconnect <device name>
```

```
[root@client ~]#nmcli networking on/off
```

```
[root@client ~]#nmcli connection show (list all connections applied on which interface)
```

```
[root@client ~]# nmcli con mod ens33 connection.autoconnect no (Turn off auto connection)
```

```
[root@client ~]# nmcli con mod ens33 ipv4.dns 8.8.8.8 (Configure a DNS server)
```

```
[root@client ~]# nmcli con mod ens33 +ipv4.dns 4.2.2.3 (Add another DNS server)
```

```
[root@client ~]# nmcli con mod ens33 +ipv4.address 10.0.0.1/24 (Add a secondary address)
```

```
[root@client ~]# nmcli con mod ens33 ipv4.method manual ipv4.address 192.168.1.10/24 ipv4.gateway  
192.168.1.1 ipv4.dns 8.8.8.8
```

Adding another connection

```
[root@client ~]# nmcli con add type ethernet con-name lab ifname ens33 ipv4.method manual  
ipv4.addresses 192.168.4.61/24 ipv4.gateway 192.168.4.2 ipv4.dns 192.168.4.2 +ipv4.dns 8.8.8.8
```

```
[root@client ~]# ifup lab
```

NOTE THAT THERE'S ONE CONNECTION ACTIVE PER TIME, IF YOU UP lab CONNECTION THE ens33 CONNECTION WILL BE NOT ACTIVE.

Editing Network Configuration

```
[root@client ~]# vim /etc/sysconfig/network-scripts/ifcfg-ens33
```

Edit in values

```
NAME=ens33  
DEVICE=ens3  
ONBOOT=yes  
NETBOOT="yes"  
UUID=6fdb13a0-1592-4992-a09e-632c23fb0d0f  
IPV6INIT=yes  
BOOTPROTO=none  
TYPE=Ethernet  
PROXY_METHOD=none  
BROWSER_ONLY=no  
DEFROUTE=yes  
IPV4_FAILURE_FATAL=no  
IPV6_AUTOCONF=yes  
IPV6_DEFROUTE=yes  
IPV6_FAILURE_FATAL=no  
IPADDR=192.168.122.20  
PREFIX=24  
GATEWAY=192.168.122.1  
DNS1=192.168.2.254
```

Restart network service on RHEL 8: **systemctl restart NetworkManager** OR **sudo nmcli connection reload**

Add Connection with Files :

```
#cd /etc/sysconfig/network-scripts/
```

```
#cp ifcfg-ens33 ifcfg-lab
```

#vi ifcfg-lab

Edit as we mention and don't forget to restart the service

Configuring Host Names:

[root@client ~]# hostname

[root@client ~]# hostname master.redhat.com

[root@client ~]# vim /etc/hostname

[root@client ~]# hostnamectl (Display information about the system)

[root@client ~]# hostnamectl status (Display information about the system)

[root@client ~]# hostnamectl set-hostname www.redhat.com

- The static host name is stored in /etc/hostname. Previous versions of RHEL stored the hostname as a variable in the /etc/sysconfig/network file.

Configure Routing:

[root@client ~]# route -n (Display routing table)

[root@client ~]# ip route (Display routing table)

[root@client ~]# ip route show (Display routing table)

[root@client ~]# ip route add 20.0.0.0/8 via 192.168.1.1 (Add a static route)

Network tools:

#netstat -tulpn

#ss -lt

#nslookup

#dig google.com

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